Infinite Sequences and Series





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11.2 Series

Infinite Series

One important application of infinite sequences is in representing "infinite summations."

Informally, if $\{a_n\}$ is an infinite sequence, then

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

Infinite series

is an **infinite series** (or simply a **series**).

The numbers a_1, a_2, a_3, are the **terms** of the series. For some series it is convenient to begin the index at $n = 0$ (or some other integer). As a typesetting convention, it is common to represent an infinite series as simply $\sum a_n$.

Convergence/Divergence

- The Nth partial sum S_N is the finite sum of the first terms.
- $\sum_{n=1}^{\infty} a_n$ converges to S if $\lim_{N \to \infty} S_N = S$ and we write:

$$S = \sum_{n=1}^{\infty} a_n$$

- If the sequence of partial sums diverges, we say that $\sum_{n=1}^{\infty} a_n$ diverges.

Geometric sequences and series

Definition:

a sequence whose consecutive terms have a common *ratio*.

i.e.

The sequence is geometric if $\frac{each\ term}{previous\ term}$=r

r is the common ratio.

Example:

1,2,4,8,……is an example of geometric sequence series.

Finding the nth term of a Geometric Sequence

$$a_n = a_1 r^{n-1}$$

$$r = \frac{a_2}{a_1}$$



Example:

Find the 15th term of the geometric sequence whose first term is 20 and whose common ratio is 1.05





Solution:

$$a_n = a_1 r^{n-1}$$
$$a_{15} = (20)(1.05)^{15-1}$$
$$a_{15} = 39.599$$



Example:

Find a formula for the nth term.

5, 15, 45, …







Solution:

$$a_n = a_1 r^{n-1}$$

$$a_n = 5(3)^{n-1}$$



Series

4 The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots$$

is convergent if $|r| < 1$ and its sum is

$$S = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1 - r} \qquad |r| < 1$$

If $|r| \geq 1$, the geometric series is divergent.



1, 4, 7, 10, 13, ….	Infinite Arithmetic	No Sum
3, 7, 11, …, 51	Finite Arithmetic	$S_n = \frac{n}{2}(a_1 + a_n)$
1, 2, 4, …, 64	Finite Geometric	$S_n = \frac{a_1(r^n - 1)}{r - 1}$
1, 2, 4, 8, …	Infinite Geometric $r \geq 1$ $r \leq -1$	No Sum
$3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots$	Infinite Geometric $-1 < r < 1$	$S = \frac{a_1}{1 - r}$

Example:

Find the sum, if possible:

$$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+...$$

Solution:



$$r = \frac{\frac{1}{2}}{1} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} \rightarrow -1 \leq r \leq 1 \rightarrow \text{Yes}$$



$$S = \frac{a_1}{1-r} = \frac{1}{1-\frac{1}{2}} = 2$$





Example: Find $\sum_{n=1}^{\infty} \frac{\pi^n}{3^{n+2}}$

Example:

Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4}$ diverges.







Example:

Find the sum, if possible:

$$\frac{2}{7} + \frac{4}{7} + \frac{8}{7} + \ldots$$









Solution:

$$r = \frac{\frac{4}{7}}{\frac{2}{7}} = \frac{\frac{8}{7}}{\frac{4}{7}} = 2 \quad \rightarrow -1 \leq r \leq 1 \rightarrow \text{No}$$

NO SUM






Series

In general, if we try to add the terms of an infinite sequence $\{a_n\}_{n=1}^{\infty}$ we get an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

which is called an **infinite series** (or just a **series**) and is denoted, for short, by the symbol

$$\sum_{n=1}^{\infty} a_n \qquad \text{or} \qquad \sum a_n$$





Series

We consider the **partial sums**

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$s_3 = a_1 + a_2 + a_3$$
$$s_4 = a_1 + a_2 + a_3 + a_4$$



and, in general,

$$s_n = a_1 + a_2 + a_3 + \cdots + a_n = \sum_{i=1}^{n} a_i$$

These partial sums form a new sequence $\{s_n\}$, which may or may not have a limit.

Example

Calculate the sum of the series $\sum_{n=1}^{\infty} a_n$ whose partial sums $s_n = \frac{n^2 - 1}{4n^2 + 1}$







Series

2 Definition Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$, let s_n denote its nth partial sum:

$$s_n = \sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$$

If the sequence $\{s_n\}$ is convergent and $\lim_{n \to \infty} s_n = s$ exists as a real number, then the series $\sum a_n$ is called **convergent** and we write

$$a_1 + a_2 + \cdots + a_n + \cdots = s \qquad \text{or} \qquad \sum_{n=1}^{\infty} a_n = s$$

The number s is called the **sum** of the series. If the sequence $\{s_n\}$ is divergent, then the series is called **divergent**.

Series

Thus the sum of a series is the limit of the sequence of partial sums.

So when we write $\sum_{n=1}^{\infty} a_n = s$, we mean that by adding sufficiently many terms of the series we can get as close as we like to the number s.

Notice that

$$\sum_{n=1}^{\infty} a_n = \lim_{n \to \infty} \sum_{i=1}^{n} a_i$$





Example: (Telescoping series)

Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent, and find its sum.



Divergence Test

If $\lim_{n\to\infty} a_n \neq 0$ then the series $\sum_{n=1}^{\infty} a_n$ diverges.

Examples: Determine whether the series is convergent or divergent. If it is convergent, find its sum.

$$\sum_{n=1}^{\infty} \frac{3n+1}{5n-2}$$

The Divergence test:

if $\sum_{n=1}^{\infty} a_n$ is a series such that

1. $\lim_{n\to\infty} a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ is div.
2. $\lim_{n\to\infty} a_n = 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ may be converge and may be divergence to decide that you should onther test.

Ex: ① $\sum_{n=1}^{\infty} n$

② $\sum_{n=1}^{\infty} \frac{2n^2+4}{n^2+2}$

Ex: $\sum_{n=1}^{\infty} \frac{1}{n}$

Sol:

Example

Show that the **harmonic series**

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is divergent.

Solution:

For this particular series it's convenient to consider the partial sums s_2, s_4, s_8, s_{16}, s_{32}, . . . and show that they become large.

$$s_2 = 1 + \tfrac{1}{2}$$

$$s_4 = 1 + \tfrac{1}{2} + \left(\tfrac{1}{3} + \tfrac{1}{4}\right) > 1 + \tfrac{1}{2} + \left(\tfrac{1}{4} + \tfrac{1}{4}\right) = 1 + \tfrac{2}{2}$$

Definition:

a **telescoping series** is a series whose partial sums eventually only have a fixed number of terms after cancellation.[1][2] The cancellation technique, with part of each term cancelling with part of the next term, is known as the **method of differences**.
For example, the series





Example 8 – *Solution*

cont'd

$$s_8 = 1 + \tfrac{1}{2} + \left(\tfrac{1}{3} + \tfrac{1}{4}\right) + \left(\tfrac{1}{5} + \tfrac{1}{6} + \tfrac{1}{7} + \tfrac{1}{8}\right)$$

$$> 1 + \tfrac{1}{2} + \left(\tfrac{1}{4} + \tfrac{1}{4}\right) + \left(\tfrac{1}{8} + \tfrac{1}{8} + \tfrac{1}{8} + \tfrac{1}{8}\right)$$

$$= 1 + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} = 1 + \tfrac{3}{2}$$

$$s_{16} = 1 + \tfrac{1}{2} + \left(\tfrac{1}{3} + \tfrac{1}{4}\right) + \left(\tfrac{1}{5} + \cdots + \tfrac{1}{8}\right) + \left(\tfrac{1}{9} + \cdots + \tfrac{1}{16}\right)$$

$$> 1 + \tfrac{1}{2} + \left(\tfrac{1}{4} + \tfrac{1}{4}\right) + \left(\tfrac{1}{8} + \cdots + \tfrac{1}{8}\right) + \left(\tfrac{1}{16} + \cdots + \tfrac{1}{16}\right)$$

$$= 1 + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2}$$

$$= 1 + \tfrac{4}{2}$$

Example 8 – *Solution*

cont'd

Similarly, $s_{32} > 1 + \frac{5}{2}$, $s_{64} > 1 + \frac{6}{2}$, and in general

$$s_{2^n} > 1 + \frac{n}{2}$$

This shows that $s_{2^n} \to \infty$ as $n \to \infty$ and so $\{s_n\}$ is divergent.

Therefore the harmonic series diverges.

Example:

Find the sum of the series $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+1)} + \frac{1}{2^n} \right)$.







Series

8 Theorem If $\Sigma\, a_n$ and $\Sigma\, b_n$ are convergent series, then so are the series $\Sigma\, ca_n$ (where c is a constant), $\Sigma\,(a_n + b_n)$, and $\Sigma\,(a_n - b_n)$, and

(i) $\displaystyle\sum_{n=1}^{\infty} ca_n = c\sum_{n=1}^{\infty} a_n$

(ii) $\displaystyle\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$

(iii) $\displaystyle\sum_{n=1}^{\infty} (a_n - b_n) = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$




